Node anonymity in networks:

The infectiousness of uniqueness

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Sunbelt | 29th of June 2023

POPNET

Universiteit van Amsterdam

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Overview

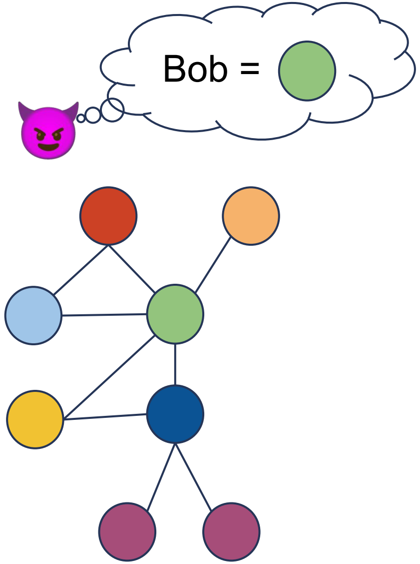
- What is anonymity in networks?
  - *k*-anonymity
- How to measure anonymity in networks?
  - *d*-*k*-Anonymity
  - Anonymity-cascade
  - Twin nodes
- How anonymous are people in a network?

Anonymity in networks

Goal: "unlocking network data"

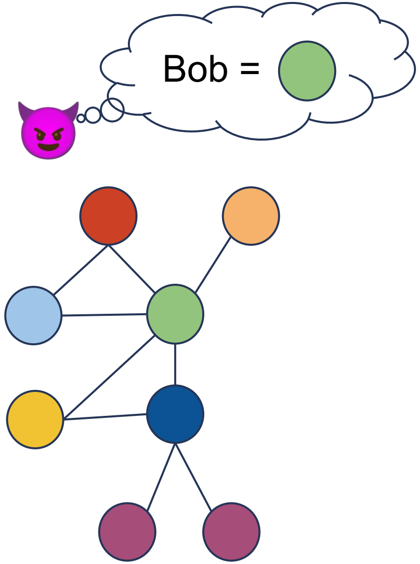
- Sensitive information
  - **Relations**
  - Attributes
- Comply with privacy laws
  - Identifiability of individuals
  - Information about individuals
- Trade off
  - Privacy vs. data utility

Anonymity in networks



Original network

Anonymity in networks



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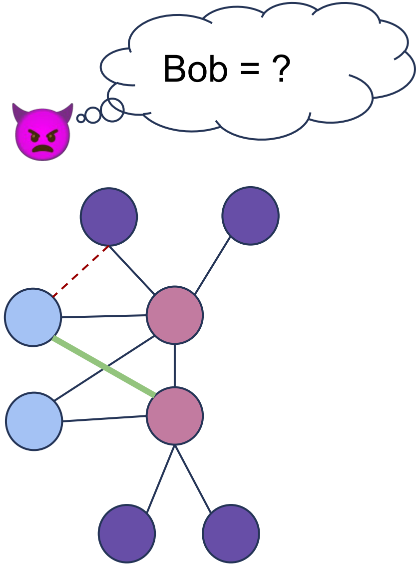
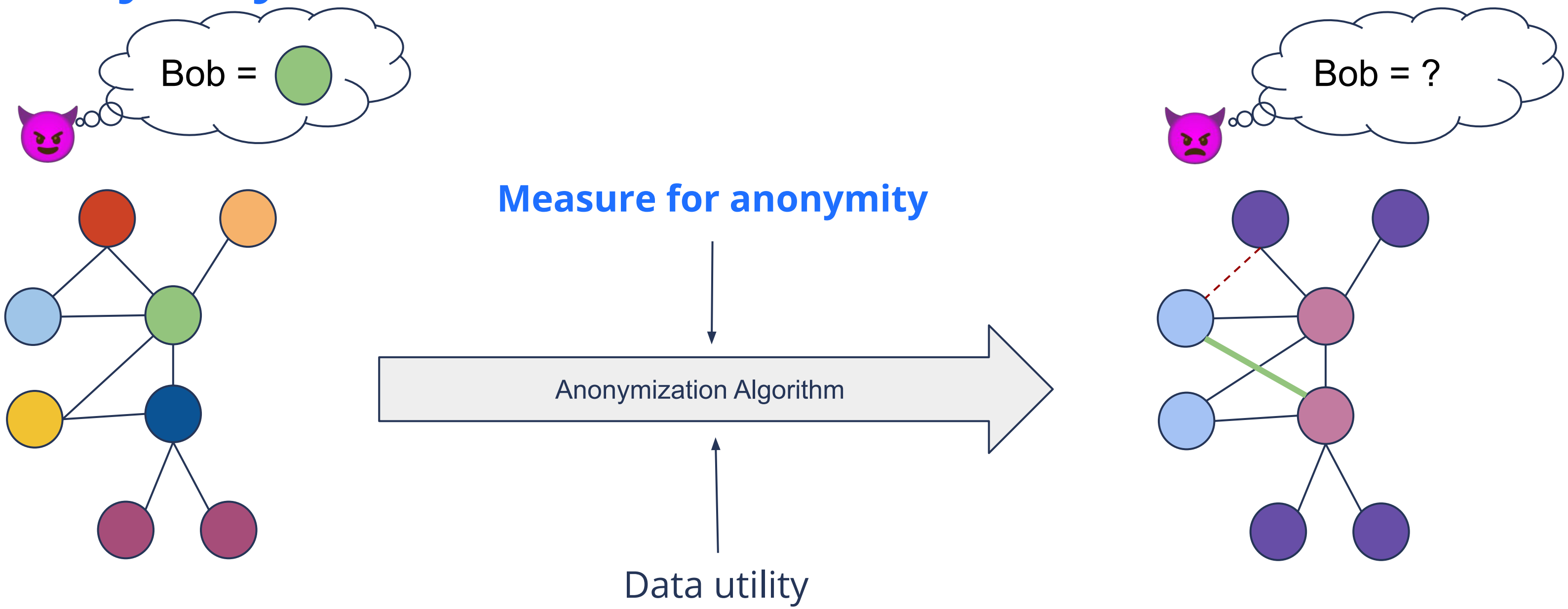


Anonymized network

Anonymity in networks



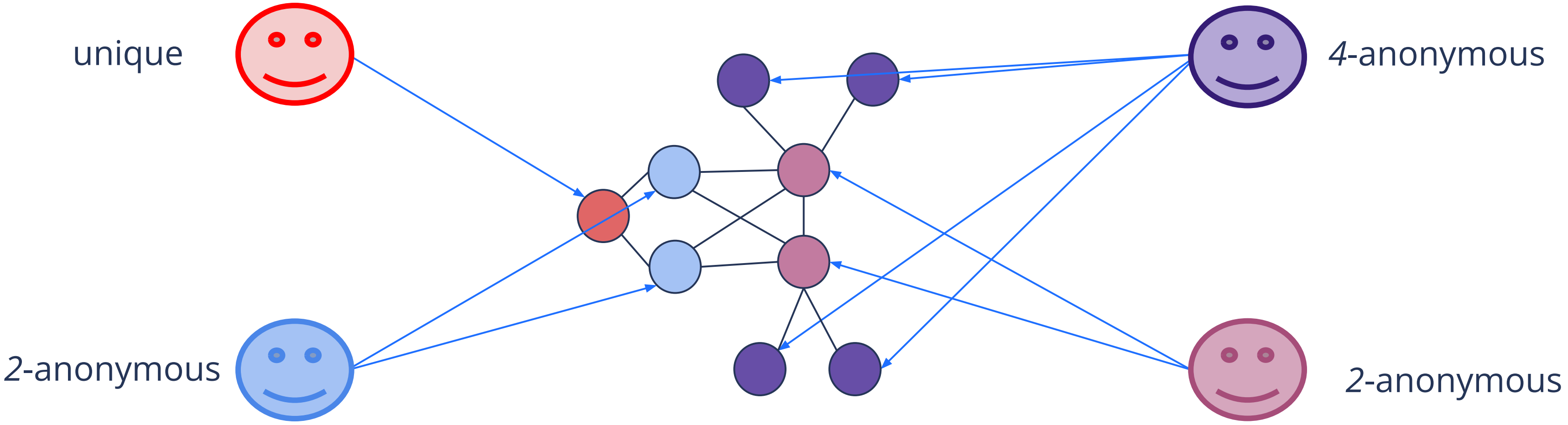
Original network

Anonymized network

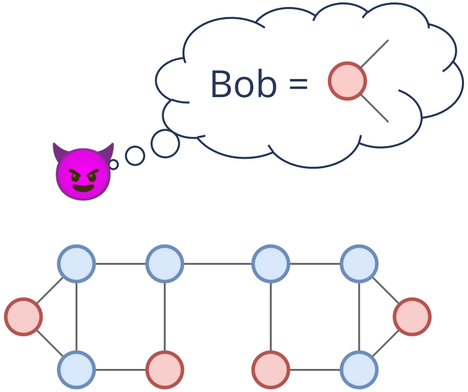
k-Anonymity

A node is *k*-anonymous if there are *k-1* equivalent nodes

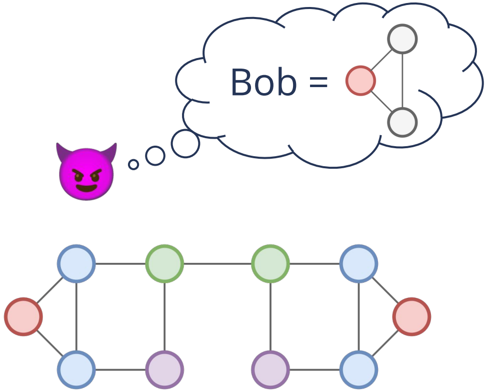
For now: anonymous = non-unique (*2*-anonymous)



Measuring anonymity



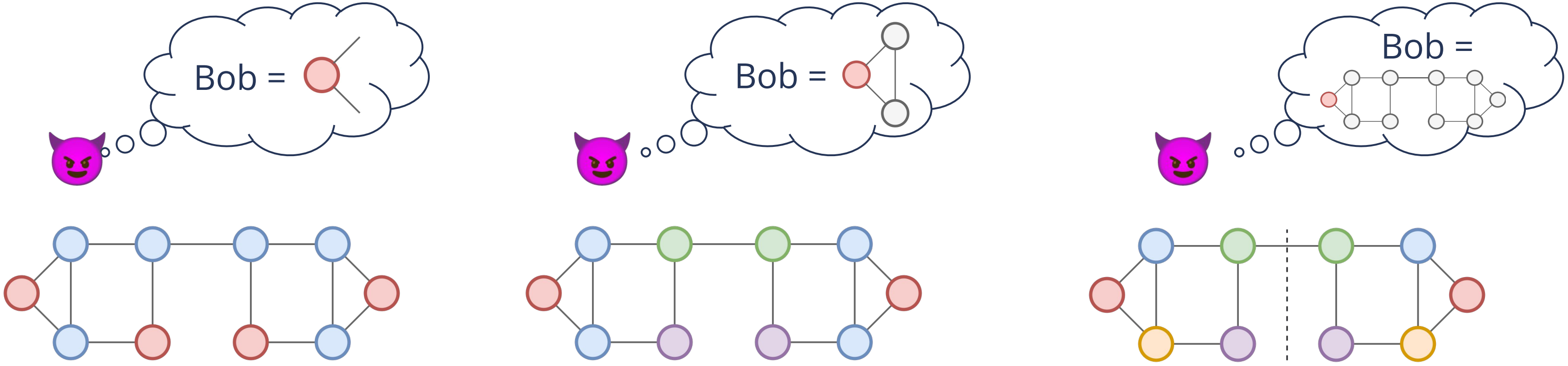
Degree



1-Neighbourhood

More strict

Measuring anonymity



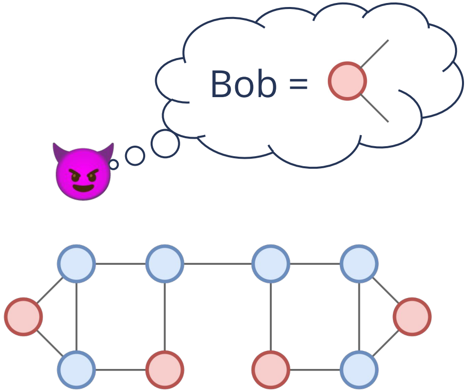
Degree

1-Neighbourhood

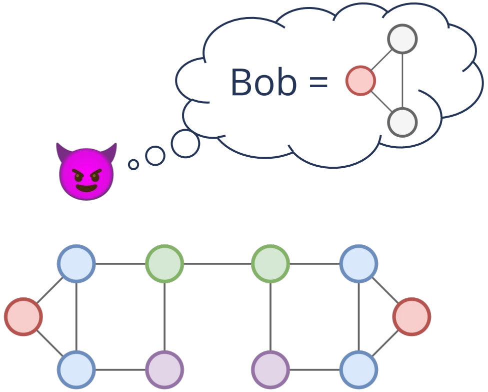
Automorphism

More strict

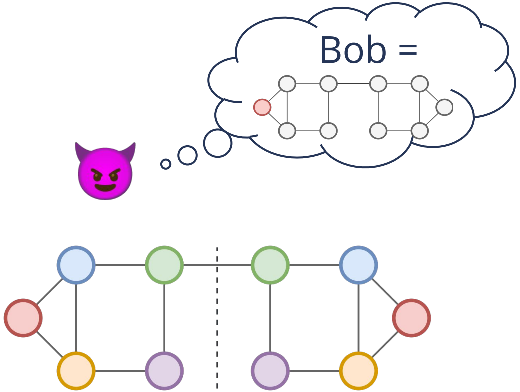
Measuring anonymity



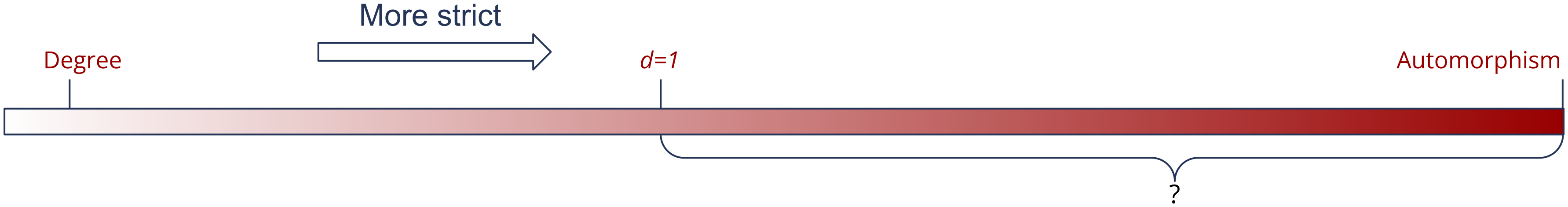
Degree



1-Neighbourhood



Automorphism



Measuring node anonymity in networks

1. *d*-*k*-Anonymity
2. Anonymity-cascade
3. Twin nodes

Q1: Is information about the ego-network enough?

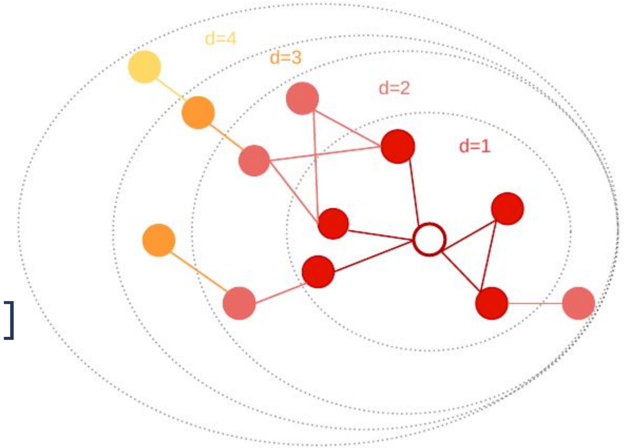
Q2: Is plain neighborhood measurement sufficient?

Approach 1: *d*-*k*-Anonymity

Nodes are equivalent if:

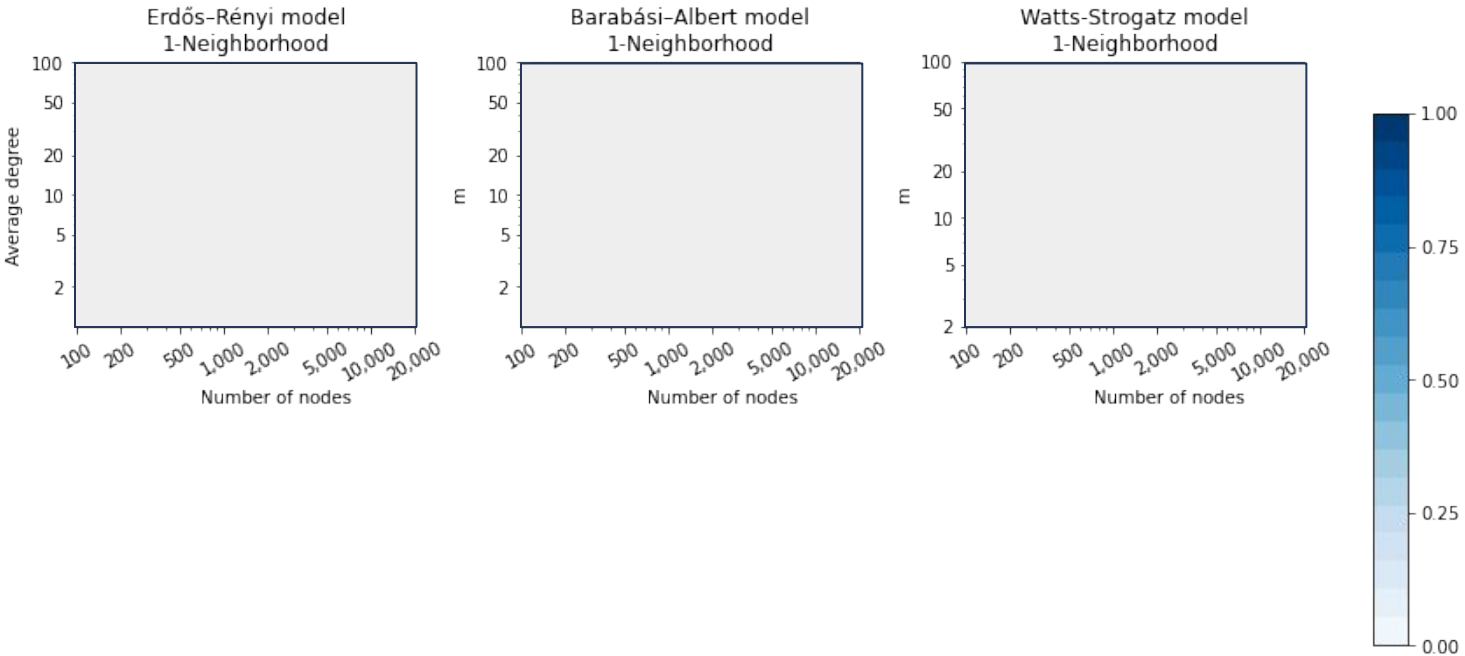
- *d*-Neighborhoods:
  - Are the same (*isomorphism*)
  - Nodes have the same structural position (*orbit*)

Computed using heuristics, iterative approach, etc. [1]



[1] de Jong, R. G., van der Loo, M. P. J. & Takes, F. W. Algorithms for efficiently computing structural anonymity in complex networks. ACM J. Exp. Algorithmics DOI: https://doi.org/10.1145/3604908 (2023).

1. *d*-*k*-Anonymity (1/2)

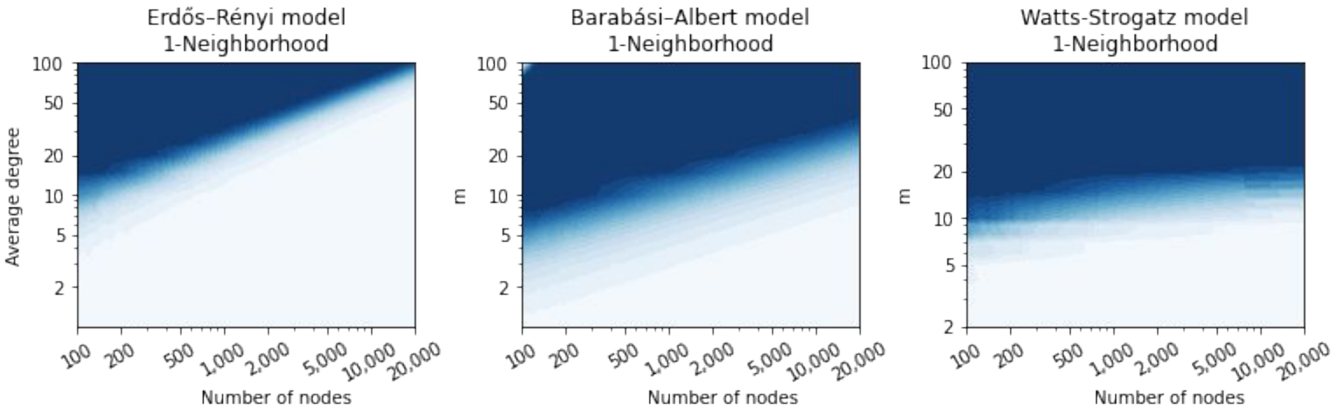


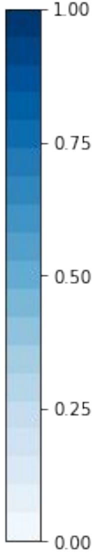
All unique 😦

$\frac{\text{\#unique nodes}}{\text{\#nodes}}$

All anonymous 😀

1. *d*-*k*-Anonymity (1/2)





All unique 😦

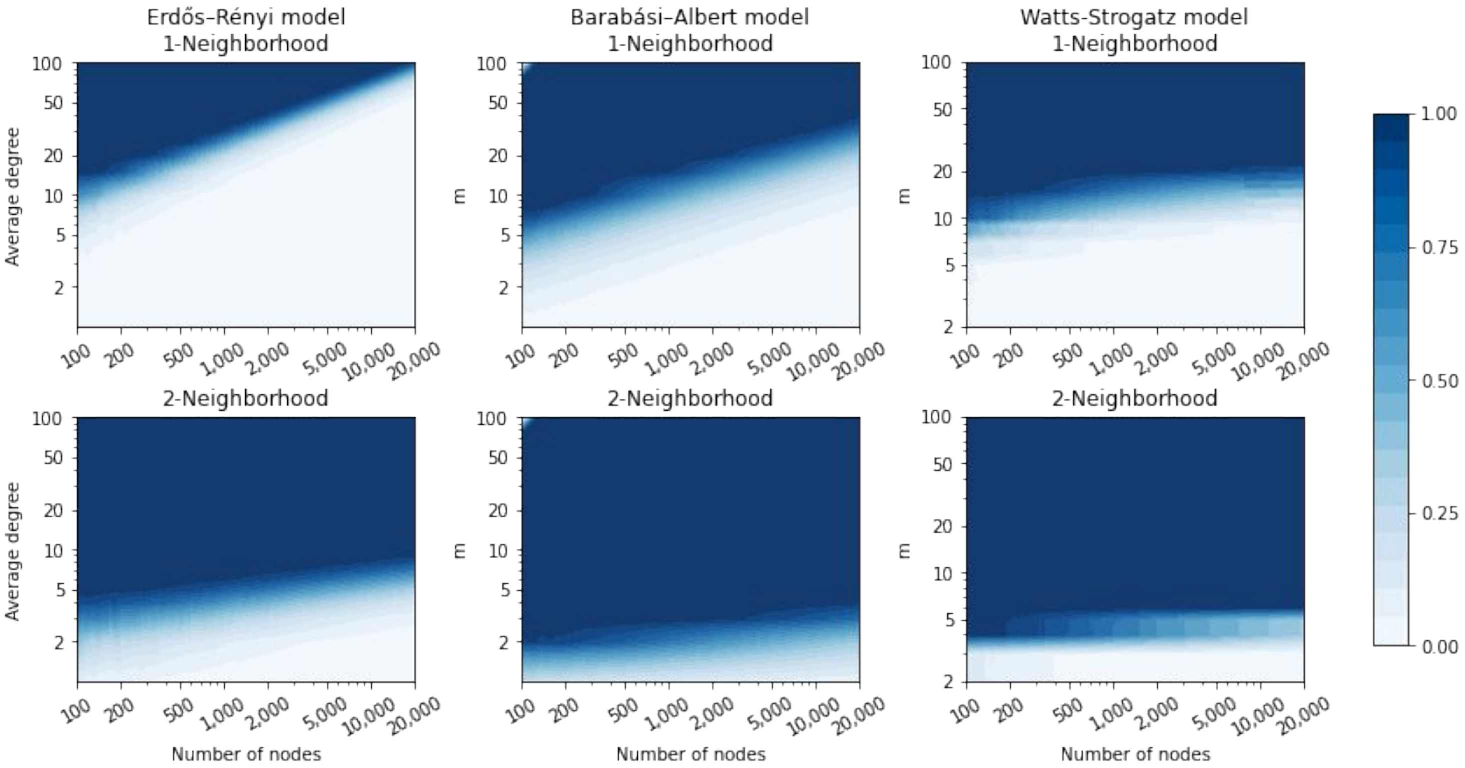
$$\frac{\#\text{unique nodes}}{\#\text{nodes}}$$

All anonymous 😀

Higher degree → less anonymity
More nodes → more anonymity
Same results as [2]

[2] Romanini, Daniele, Sune Lehmann, and Mikko Kivelä. "Privacy and uniqueness of neighborhoods in social networks." Scientific reports 11.1 (2021): 20104.

1. *d*-*k*-Anonymity (1/2)



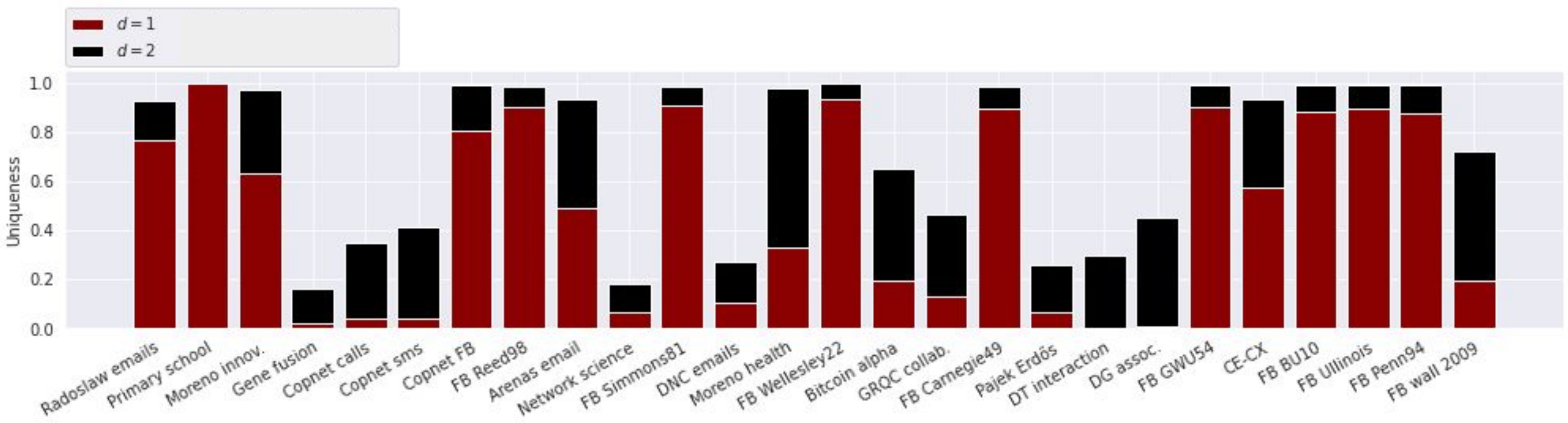
All unique 😦

$\frac{\#\text{unique nodes}}{\#\text{nodes}}$

All anonymous 😀

Almost no anonymity for d=2. Size has small / no effect

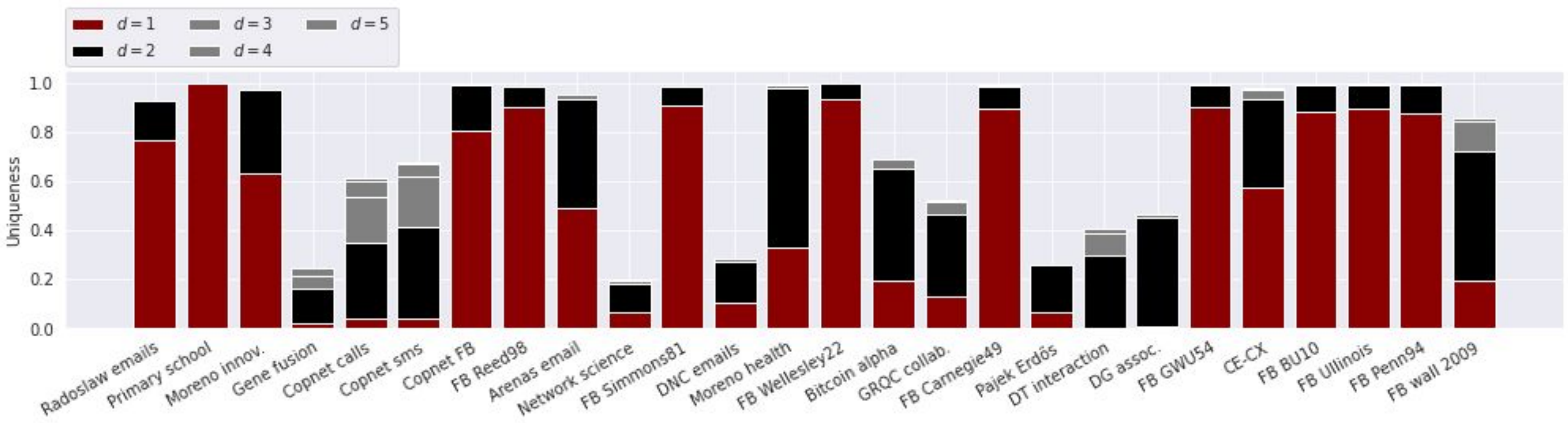
d-*k*-Anonymity (2/2)



|V|: 167 - 46,952 |E|: 279 - 1,362,220

Big decrease anonymity d=1 → d=2

d-*k*-Anonymity (2/2)



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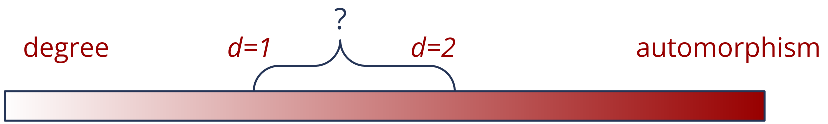
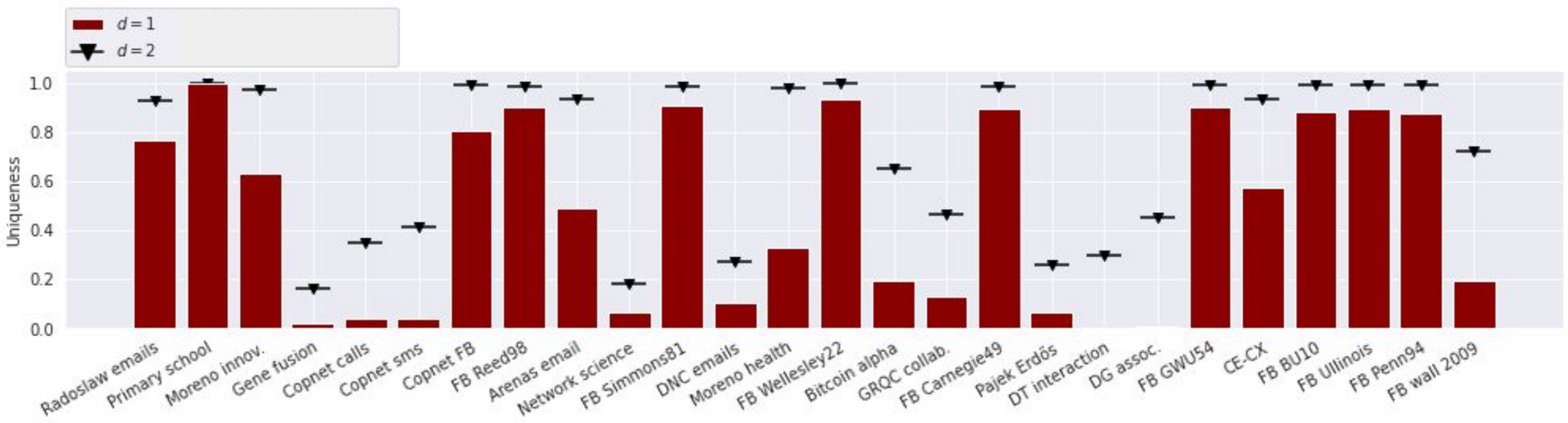
Big decrease anonymity d=1 → d=2

Overall small effect d=2 → d>2



d-*k*-Anonymity



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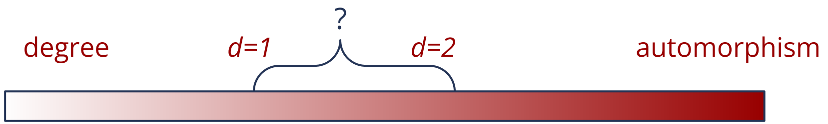
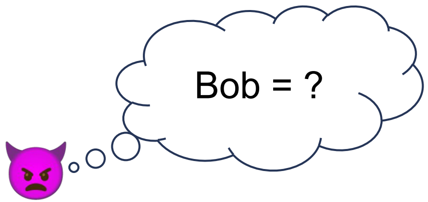
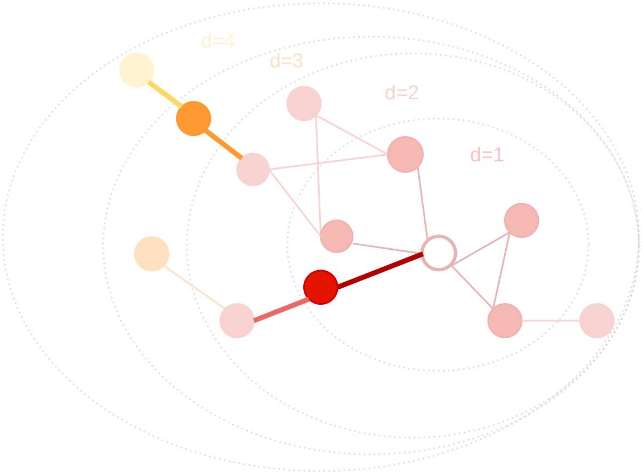


Approach 2: Anonymity-cascade



Knowledge:

- *1*-neighborhood



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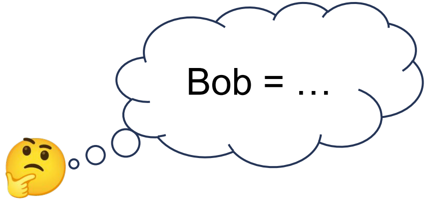
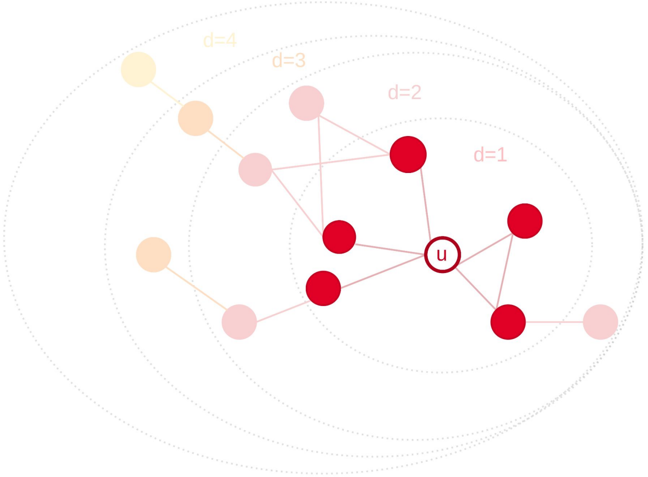


Knowledge:

- *1*-neighborhood
- **Existence of a connection to unique node**



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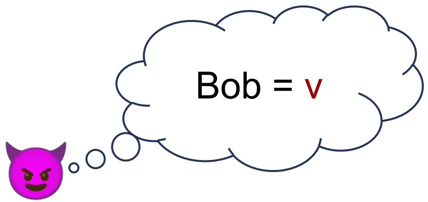
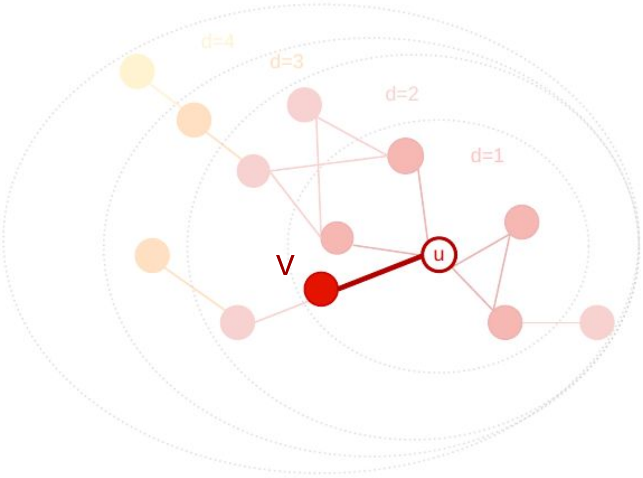


Knowledge:

- *1*-neighborhood
- **Existence of a connection to unique node**
    - *"Infecting nodes with uniqueness"*



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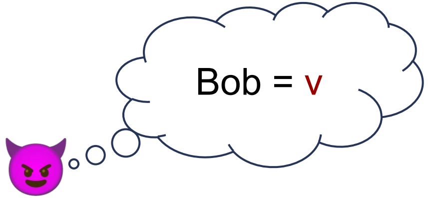
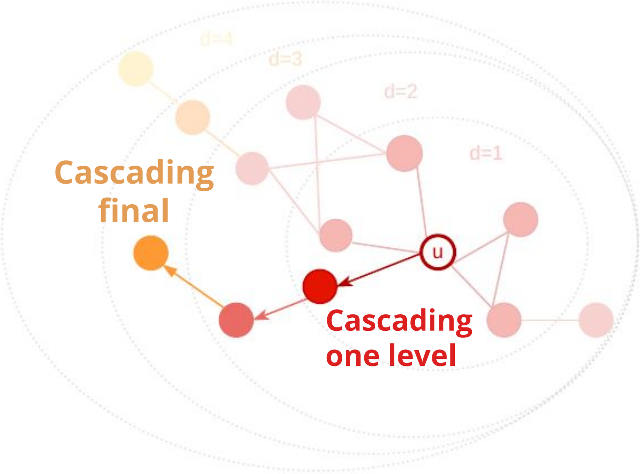
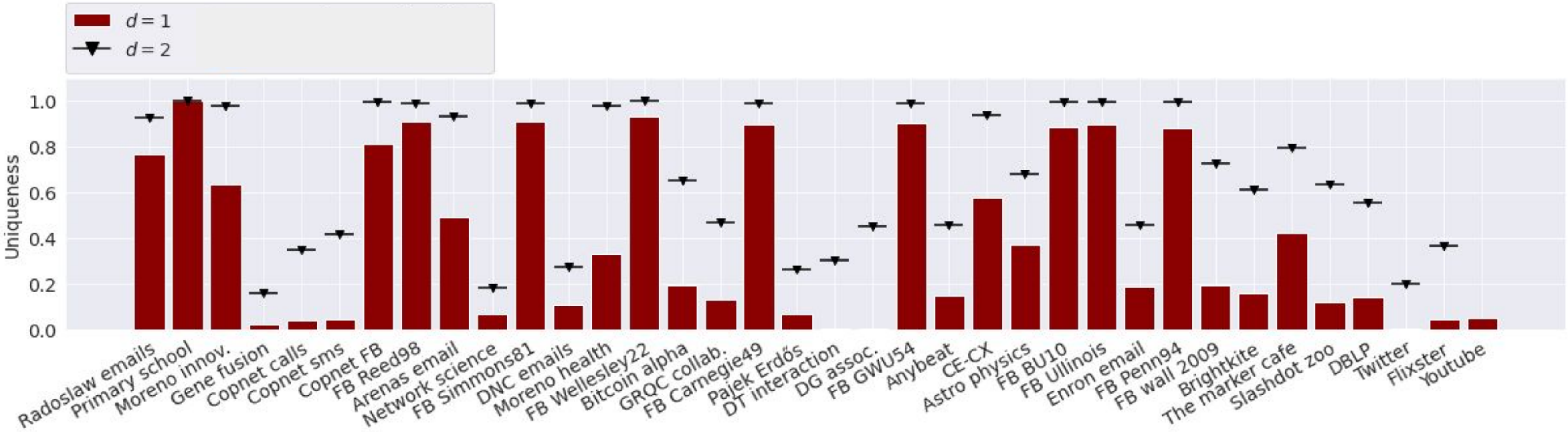


Knowledge:

- *1*-neighborhood
- **Existence of a connection to unique node**
  - *"Infecting nodes with uniqueness"*
- One cascading level ≤ *d-k-anonymity* with *d=2*



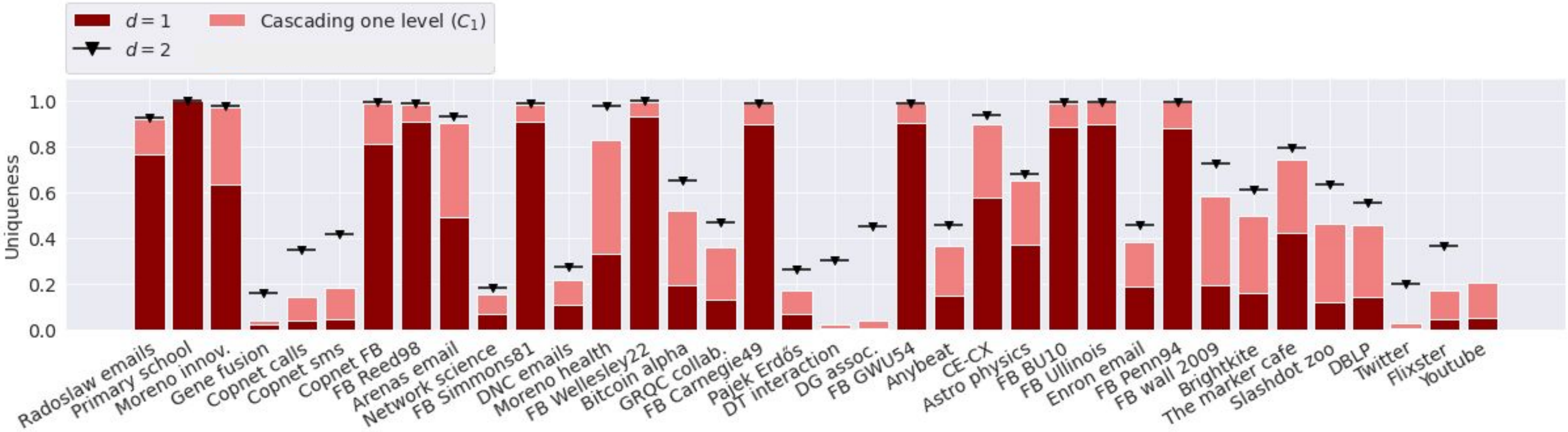
Anonymity-Cascade



|V|: 167 - 3,223,590 |E|: 279 - 18,750,747

*More networks due to lower runtimes *d*=2 and cascading

Anonymity-Cascade



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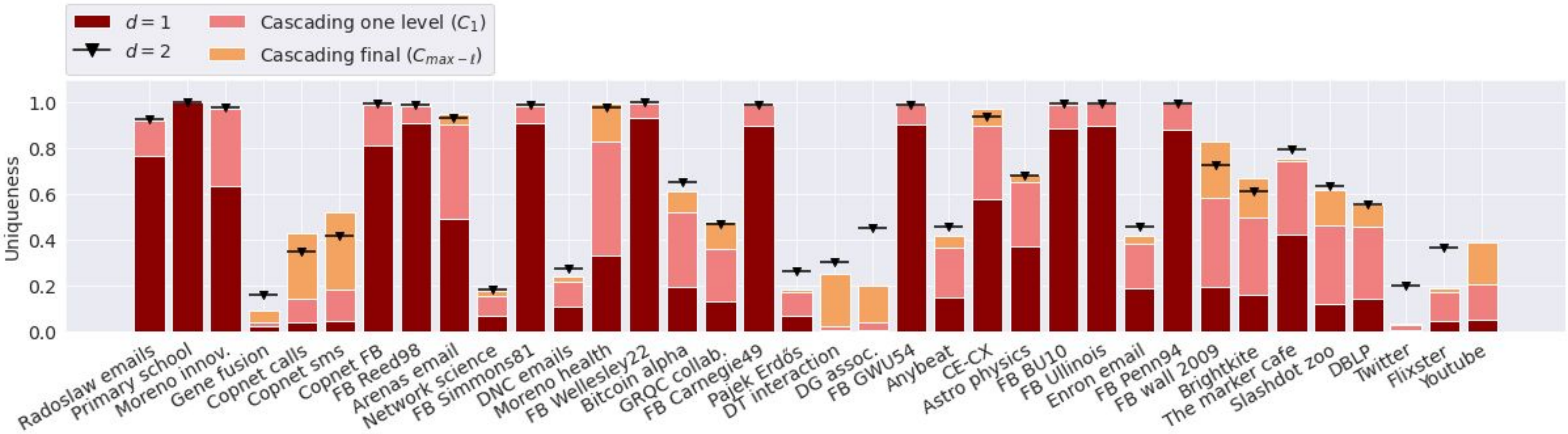
*More networks due to lower runtimes *d=2* and cascading

>100% increase: 19 / 36

Large increase in uniqueness with knowledge of one connection

Approximate *d-k*-anonymity with *d=2*

Anonymity-Cascade



|V|: 167 - 3,223,590 |E|: 279 - 18,750,747

*More networks due to lower runtimes *d*=2 and cascading

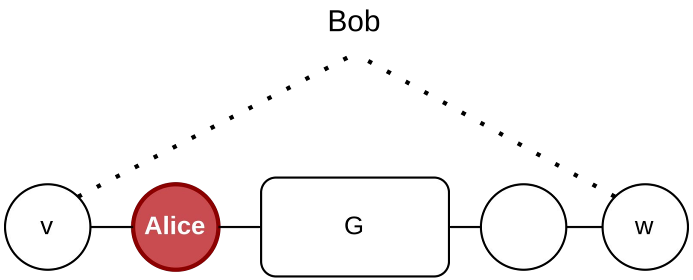
>100% increase: 19 / 5 / 36

Large increase in uniqueness with knowledge of one connection

Approximate *d*-*k*-anonymity with *d*=2

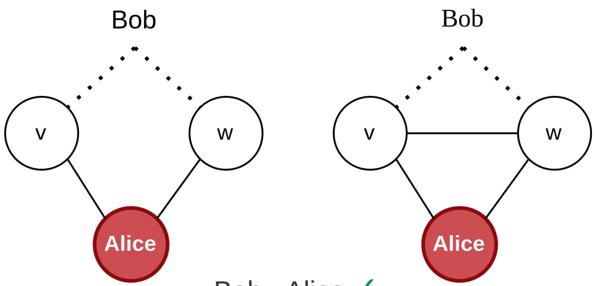
Increase for some networks with cascading final

Approach 3: Twin Nodes



Bob - Alice ?

v and *w* structurally indistinguishable
(same orbit)



Bob - Alice ✓

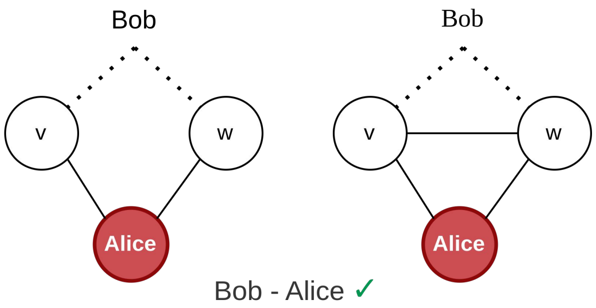
Open twins:
$N_1(v) - \{v\} = N_1(w) - \{w\}$

Closed twins:
$N_1(v) = N_1(w)$

→ *v*, *w* structurally indistinguishable

Approach 3: Twin Nodes

- Twin nodes share their connections
  - Same degree
  - Same 1-neighborhood
  - Same 2-neighborhood
  - ...
  - Structurally indistinguishable!



Open twins: $N_1(v) - \{v\} = N_1(w) - \{w\}$

Closed twins: $N_1(v) = N_1(w)$

→ *v*, *w* structurally indistinguishable

Network	Nodes	Edges	Fraction twins	$max-\ell$	Runtime *d-k-anonymity* d=1	d=2	Runtime *Anonymity-cascade* C_1	$C_{max-\ell}$	Type
Radoslaw emails[28]	167	3,250	0.072	3	0.02 s	+ < 0.01 s	+ < 0.01 s	+ < 0.01 s	Communication
Primary school[29]	236	5,899	0.000	1	0.02 s	+ < 0.01 s	+ < 0.01 s	+ < 0.01 s	Human contact
Moreno innov.[28]	241	923	0.025	3	< 0.01 s	+ < 0.01 s	+ < 0.01 s	+ < 0.01 s	Communication
Gene fusion[28]	291	279	0.753	6	< 0.01 s	+ < 0.01 s	+ < 0.01 s	+ < 0.01 s	Biological
Copnet calls[3]	536	621	0.287	10	< 0.01 s	+ 0.01 s	+ < 0.01 s	+ < 0.01 s	Communication
Copnet sms[3]	568	697	0.285	7	< 0.01 s	+ 0.01 s	+ < 0.01 s	+ < 0.01 s	Communication
Copnet FB[3]	800	6,418	0.005	4	0.02 s	+ 0.01 s	+ < 0.01 s	+ < 0.01 s	Online social
FB Reed98[30]	962	18,812	0.012	3	0.05 s	+ 0.01 s	+ 0.01 s	+ 0.02 s	Online social
Arenas email[28]	1,133	5,451	0.042	5	0.02 s	+ 0.01 s	+ < 0.01 s	+ < 0.01 s	Communication
Network science[28]	1,461	2,742	0.755	6	0.01 s	+ 0.01 s	+ < 0.01 s	+ < 0.01 s	Co-autorship
FB Simmons81[30]	1,518	32,988	0.011	3	0.12 s	+ 0.02 s	+ 0.01 s	+ 0.02 s	Online social
DNC emails[28]	1,893	4,385	0.706	4	0.02 s	+ 0.38 s	+ < 0.01 s	+ < 0.01 s	Online social
Moreno health[28]	2,539	10,455	0.003	5	0.03 s	+ 0.04 s	+ < 0.01 s	+ < 0.01 s	Human social
FB Wellesley22[30]	2,970	94,899	0.000	4	0.4 s	+ 0.07 s	+ 0.04 s	+ 0.08 s	Online social
Bitcoin alpha[30]	3,783	14,124	0.306	5	0.04 s	+ 0.36 s	+ < 0.01 s	+ 0.01 s	Online social (trust)
GRQC collab.[31]	5,242	14,484	0.455	8	0.03 s	+ 0.04 s	+ < 0.01 s	+ < 0.01 s	Co-autorship
FB Carnegie49[30]	6,637	249,967	0.008	4	1.37 s	+ 0.44 s	+ 0.09 s	+ 0.18 s	Online social
Pajek Erdős[28]	6,927	11,850	0.737	6	0.02 s	+ 0.19 s	+ 0.01 s	+ 0.02 s	Co-autorship
DT interaction[32]	7,341	15,138	0.572	12	0.07 s	+ 2.88 s	+ < 0.01 s	+ < 0.01 s	Biological
DG assoc.[32]	7,813	21,357	0.531	8	0.21 s	+ 5.13 s	+ < 0.01 s	+ < 0.01 s	Biological
FB GWU54[30]	12,193	469,528	0.006	4	2.64 s	+ 0.92 s	+ 0.18 s	+ 0.36 s	Online social
Anybeat[30]	12,645	49,132	0.500	5	0.41 s	+ 1.34 h	+ 0.02 s	+ 0.04 s	Online social
CE-CX[30]	15,229	245,952	0.021	6	1.04 s	+ 1.56 s	+ 0.09 s	+ 0.19 s	Biological
Astro Physics[30]	18,771	198,050	0.305	6	0.72 s	+ 1.91 s	+ 0.05 s	+ 0.11 s	Co-autorship
FB BU10[30]	19,700	637,528	0.006	4	3.17 s	+ 1.44 s	+ 0.25 s	+ 0.50 s	Online social
FB Uillinois[30]	30,664	1,048,574	0.002	4	5.96 s	+ 2.66 s	+ 0.39 s	+ 0.78 s	Online social
Enron email[31]	36,692	183,831	0.528	6	0.90 s	+ 1.44 m	+ 0.06 s	+ 0.12 s	Communication
FB Penn94[30]	41,536	1,362,220	0.002	4	8.95 s	+ 5.39 s	+ 0.50 s	+ 1.00 s	Online social
FB wall 2009[28]	46,952	183,412	0.133	8	0.54 s	+ 1.80 s	+ 0.05 s	+ 0.13 s	Communication
Brightkite[30]	58,228	214,078	0.258	8	0.78 s	+ 18.41 s	+ 0.06 s	+ 0.14 s	Online social
The marker cafe[33]	69,413	1,644,843	0.200	5	37.96 s	+ 10.46 m	+ 0.67 s	+ 1.35 s	Human contact
Slashdot zoo[28]	79,116	467,731	0.274	7	3.36 s	+ 2.62 m	+ 0.15 s	+ 0.34 s	Online social
Twitter[28]	465,017	833,540	0.801	5	59.53 s	+ 6.04 h	+ 0.29 s	+ 0.62 s	Online social
DBLP[28]	1,824,701	8,344,615	0.402	10	51.37 s	+ 11.58 m	+ 28.1 s	+ 57.18 s	Co-autorship
Flixster[28]	2,523,386	7,918,801	0.631	8	3.10 m	+ 8.57 h	+ 18.58 s	+ 37.81 s	Online social
Youtube[28]	3,223,585	9,375,374	0.000	15	13.24 m	+ > 1 week	+ 30.68 s	+ 1.04 m	Online social

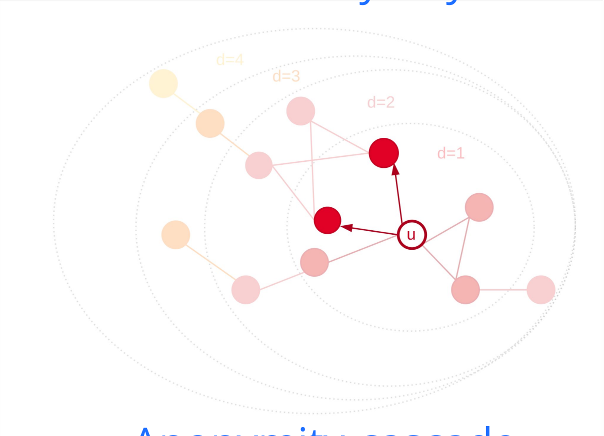
Twin nodes can occur frequently: max **0.801**

Can not be unique!

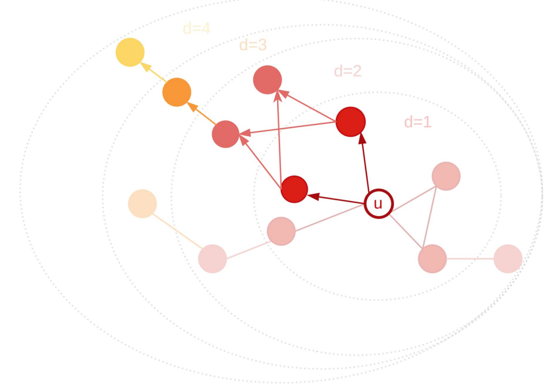
Approach 3: Twin Nodes

- Twin nodes share their connections
  - **Structurally indistinguishable**
- Twin-unique:
  - Unique
  - All candidates are twins
    - *d*-*k*-anonymity + anonymity-cascade
    - Cascade from twin-unique nodes

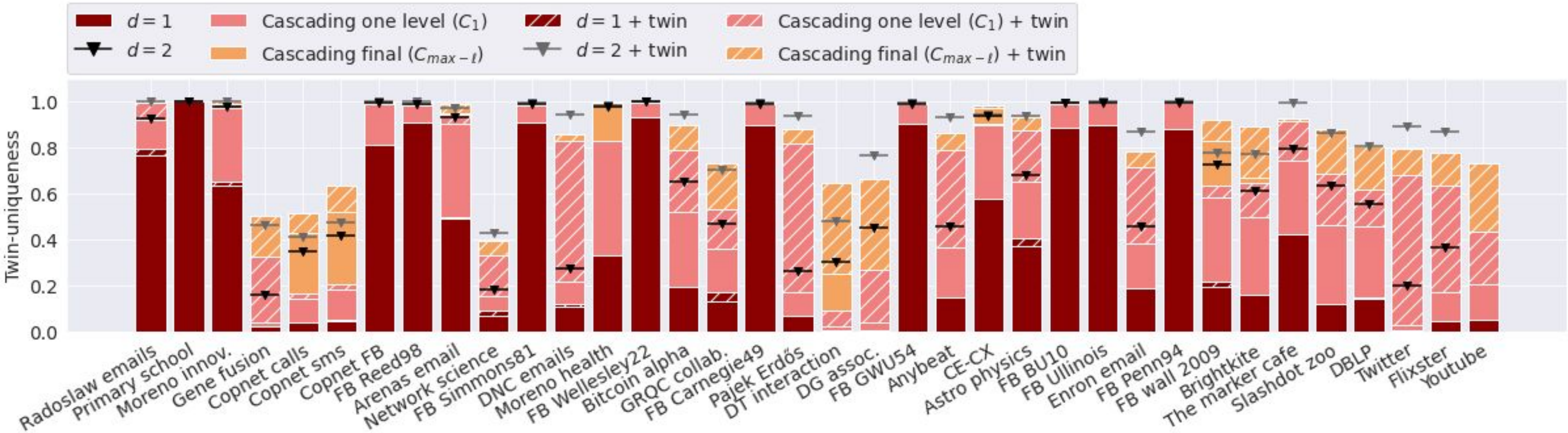
d-*k*-Anonymity



Anonymity-cascade



Twin-uniqueness



|V|: 167 - 3,223,590 |E|: 279 - 18,750,747

Large increase twin-uniqueness compared to uniqueness

Conclusions

- To assess anonymity, we must look beyond the ego network
  - Substantial difference between *d=1* and *d=2*
  - ***d-k*-Anonymity** looks beyond the direct neighborhood
- Uniqueness is infectious
  - **Anonymity-cascade** captures this effect
  - Knowledge of one extra link: large decreases in anonymity
- Twin nodes
  - Structurally indistinguishable, but occur frequently
  - Incorporating **twin-uniqueness** affects both *d*-*k*-anonymity and anonymity-cascade

Paper:
https://arxiv.org/abs/2306.13508

Thank you!

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- **Frank Takes**: The small-world structure of a population-scale social network; Thursday, June 29th at 10:40am
- **Rachel de Jong**: Node anonymity in networks: The infectiousness of uniqueness; Thursday, June 29th at 1:30pm
- **Hanjo Boekhout**: Early warning signals for predicting vendor success in dark net forum networks; Friday, June 30th at 9:00am
- **Dominika Czerniawska**: The Down of Global Scientific Collaborations? The Nuclear Fusion Community under the Shadow of the Russia's Invasion of Ukraine; Saturday, July 1st at 10:40am
- **Yuliia Kazmina**: Social-economic segregation in a Population-Scale Social Network; Saturday, July 1st at 3:10pm